

Max Boksem

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Education

Universiteit van Amsterdam

Aug 2023 – Mar 2025

Master of Science in Software Engineering (Grade: 8.2 / 10.0) (Cum Laude)

Amsterdam, Netherlands

- **Relevant Coursework:** Requirements Engineering, Software Specification, Verification and Testing, Model-based Design of Cyber-physical Systems, Software Evolution, DevOps and Cloud-based Software, Software Process
- **Thesis:** Bridging Incremental Programming and Complex Software Development in Modern Programming Environments (Grade: 8.5 / 10.0)

McGill University

Sep 2016 – May 2020

Bachelor of Science in Computer Science, minor in Physics (GPA: 3.0 / 4.0) [* = Graduate level]

Montréal, Canada

- **Relevant Coursework:** Algorithms and Data Structures, Software Design, Artificial Intelligence, Database Systems, Computer Graphics*, Computer Networks*, Applied Machine Learning*, Data Analysis and Computation (Physics), Signal Processing (Physics)

Experience

Universiteit van Amsterdam

Jan 2025 – Aug 2025

Research Engineer | Rust, React, Tokio, Node.js, Express, Python, Full-stack Development

Amsterdam, Netherlands

- Designed and implemented the CLO engine, a discrete-event world emulator combining declarative logic inference with parallel execution in Rust. Developed a custom language parser, reactive rule engine, and modular execution model inspired by Logic Programming Petri Nets (LPPNs) and BDI agent systems.
- Conducted ongoing research and full-stack development of Incremental Graph Code (IGC), a new exploratory and incremental programming environment built with React, Node.js, Express, and Electron. Enabled users to construct applications through interactive execution graphs; project outcomes resulted in two published research papers.

Cisco Systems

May 2019 – Aug 2023

Software Engineer | Django, Python, React, Golang, C, Kafka, Full-stack Development

Kanata, Canada

- Full stack development working on 6 different tools (micro-services) for build workflow automation and UAT used by all Cisco IOS-XE developers. Millions of dollars are spent through these systems for teams around the world.
- Led multiple performance optimization efforts introducing compiled C and Golang into the build process, reducing overall build times by up to 15%. Additionally, improved full-stack GUI performance across multiple internal tools, achieving 200–10,000% faster load and execution times.
- Implemented pyATS based automation which performed packet injection testing with target system CPU and memory numerical analysis to identify the specific performance profile of different Cisco switch families.

Holland America Group

Jun 2018 – Aug 2018

Software Development Intern | Python, PostgreSQL, JS/HTML/CSS, PHP, Full-stack Development

Seattle, United States

- Prepared and presented a model of the architecture of a database system and software model that involved over 30 characterizations that could contribute to incident reports.
- Developed, from scratch, software to quickly analyze attributes of incident reports on 37 different cruise ships using languages such as Python, PostgreSQL, JavaScript, HTML/CSS, and PHP.

Publications and Awards

Exploration and Complexity Management in Graph-Based Programming Environments

Apr 2025

Presented at <Programming> 2025 (PX)

Prague, Czechia

- Continuation paper expanding Incremental Graph Code (IGC) to explore how graph-based programming environments can better support exploration and complexity management. [\[DOI\]](#)

Bridging Incremental Programming and Complex Software Development Environments

Oct 2024

Presented at Splash 2024 (PAINT)

Pasadena, United States

- A paper introducing Incremental Graph Code (IGC), a hybrid graph-based programming environment demonstrating the feasibility of unifying IDE-style complexity management with REPL-style exploratory programming. [\[DOI\]](#)

Tomlinson Engagement Awardee for Mentoring (TEAM) Award ×5

2017-2020

- Recognizes dedication and excellence in peer mentorship roles through volunteering at the Computer Science Helpdesk for 5 semesters.

Technical Skills

Languages: Java, C, Python, C#, C++, Rust, Golang, HTML/CSS, JS/TS, OCaml, Haskell, Swift, x86

Technologies: React.js, Django, Flask, Express.js, jQuery, Node.js, Electron, SQL-style/Mongo DBs, Git, REST, Kafka, gRPC, Unix Systems, Full-stack Development, Multi-threading, VS Code, Visual Studio, JetBrains IDEs, Postman, Compass, Unity